Shivam Kaushik

Researcher @ University of Florida

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RESEARCH EXPERIENCE

Sep 2023 – Present	**Biological Scientist** **Lab Technician** **University of Florida**, Gainesville, USA *Supervisor:* Dr. Diego E. Rincon-Limas	*until Sep 2025*
Jan 2022 – Dec 2022	**Graduate Student** **University of North Carolina**, Chapel Hill, USA *Supervisor:* Dr. Toshi Hige	
Jun 2017 – Dec 2021	**Research Assistant** **Summer Training** **Regional Centre for Biotechnology**, India *Supervisor:* Dr. Pinky Kain	*Aug 2018 - Dec 2021* *Jun 2017 - Aug 2017*

EDUCATION

Jan 2022 – Dec 2022	**Ph.D. Researcher** | Major: ***Neuroscience*** **University of North Carolina**, NC, USA
Jul 2018 – Jul 2020	**Master of Technology** | Major: ***Biotechnology*** | GPA: *3.5/4.0* **Maharishi Dayanand University**, India Dissertation: *Mapping novel neural circuits for feeding and foraging in Drosophila melanogaster*
Jul 2014 – May 2018	**Bachelor of Technology** | Major: ***Biotechnology*** | GPA: *3.8/4.0* **Maharishi Dayanand University**, India Dissertation: *Dietary modulation of taste behavior & its impact on aging in Drosophila melanogaster*

PUBLICATIONS

Peer-Reviewed Publications

Neurosci Insights 2026	**Shivam Kaushik**, Kartik Singh, Rahul Kumar, Sachin Kumar, Srishti Sanghi, Teiichi Tanimura, Diego E. Rincon-Limas, Pinky Kain, *Activity and State-Dependent Modulation of Salt Taste Behavior via Pharyngeal Neurons in Drosophila Melanogaster*, Neuroscience Insights 2026.
Ann Neurol 2024	Swapnil Pandey, Deepak Chhangani, **Shivam Kaushik**, and Diego E. Rincon-Limas, *The microRNA mir-33 regulates TDP-43-induced neurotoxicity in a fly model of FTLD/ALS*, Annals of Neurology.
eNeuro 2023	Andrew M. Davidson, **Shivam Kaushik**, and Toshihide Hige, *Dopamine-Dependent Plasticity Is Heterogeneously Expressed by Presynaptic Calcium Activity across Individual Boutons of the Drosophila Mushroom Body*, eNeuro, 10(10), ENEURO.0275-23.2023.
J Vis Exp 2023	Manal Shakeel, **Shivam Kaushik**, Teiichi Tanimura, Axel Brockmann, and Pinky Kain, *Tracking Sugar-Elicited Local Searching Behavior in Drosophila*, Journal of Visualized Experiments, (201), e65955.
IJMS 2022	Gouri Chopra, **Shivam Kaushik**, and Pinky Kain, *Nutrient Sensing via Gut in Drosophila melanogaster*, International Journal of Molecular Sciences, 23(5), 2694.
Sci Prog 2022	**Shivam Kaushik**, Srishti Sanghi, Sachin Kumar, Rahul Kumar, and Pinky Kain, *Modulation of sugar feeding behavior by Gymnema sylvestre in Drosophila melanogaster*, Science Progress.
Alzheimers Dement 2021	**Shivam Kaushik**, Diego E. Rincon-Limas, and Pinky Kain, *Starvation and activity dependent modulation of salt taste behavior in Drosophila*, Alzheimer's & Dementia, 17(Suppl 12), e058538.
ACS Omega 2020	Sujeet Thakur, Krishnendu Goswami, Pallavi Rao, **Shivam Kaushik**, Bhanu Singh, Pinky Kain, Shailendra Asthana, Saikat Bhattacharjee, Prasenjit Guchhait, and S. Eswaran, *Fluoresceinated aminohexanol tethered inositol hexakisphosphate: Studies on Arabidopsis thaliana and Drosophila melanogaster and docking with 2P1M Receptor*, ACS Omega, 5(16), 9585–9597.
JEN 2018	**Shivam Kaushik**, Rahul Kumar, and Pinky Kain, *Salt an essential nutrient: Advances in understanding salt taste detection using Drosophila as a Model System*, Journal of Experimental Neuroscience, 12, 1179069518806894.

Publications

BioRixv 2024	**Shivam Kaushik**, Rahul Kumar, Sachin Kumar, Srishti Sanghi, Teiichi Tanimura, Diego E. Rincon-Limas, and Pinky Kain, *Activity and state-dependent modulation of salt taste behavior via pharyngeal neurons in Drosophila melanogaster*, BioRixv 2024.

Book Chapters

IntechOpen 2019	**Shivam Kaushik** and Pinky Kain, *Understanding taste using Drosophila melanogaster*, in *Animal Models in Medicine and Biology* (Eds. Eva Tvrdá and Sarat Chandra Yenisetti), IntechOpen.
IntechOpen 2020	**Shivam Kaushik**, Shivangi Rawat, and Pinky Kain, *A look into the central taste circuits using Drosophila melanogaster as a model system*, in *Role of Obesity in Human Health and Disease* (Eds. Venketeshwer Rao and Leticia Rao), IntechOpen.
IntechOpen 2021	Zoha Sadaqat, **Shivam Kaushik**, and Pinky Kain, *Gut feeding the brain: Drosophila gut an animal model for medicine to understand mechanism mediating food preferences*, in *Preclinical Animal Modeling in Medicine* (Eds. Enkhsaikhan Purevjav, Joseph F. Pierre, and Lu Lu), IntechOpen.

Manuscripts in Preparation

- **Shivam Kaushik** and Diego E. Rincon-Limas, *Optogenetic control of neurodegenerative protein aggregation in Drosophila*.
- Swapnil Pandey, **Shivam Kaushik**, Deepak Chhangani, and Diego E. Rincon-Limas, *The SARS-CoV-2 NSP1 protein promotes amyloid beta aggregation and toxicity in vivo*.

TALKS & POSTERS

FL Genetics 2025	James Caprio, Riana Shah, Camila Campos, Deepak Chhangani, **Shivam Kaushik**, Swapnil Pandey, and Diego E. Rincon-Limas, *Investigating the Impact of SARS-CoV-2 Proteins on Neurodegeneration in Fruit Flies*, Florida Genetics Symposium by University of Florida, Gainesville, Florida, USA. November 5–6, 2025.
FL Genetics 2025	Arushi Pant, Jessica Jiang, Deepak Chhangani, **Shivam Kaushik**, Swapnil Pandey, and Diego E. Rincon-Limas, *Drosophila models of Amyotrophic lateral sclerosis (ALS)*, Florida Genetics Symposium by University of Florida, Gainesville, Florida, USA. November 5–6, 2025.
UF URS 2025	James Caprio, Riana Shah, Camila Campos, Deepak Chhangani, **Shivam Kaushik**, Swapnil Pandey, and Diego E. Rincon-Limas, *Studying the role of SARS-CoV-2 proteins in neurodegeneration using fruit flies*, Undergraduate Research Symposium organized by University of Florida, Gainesville, Florida, USA. October 16, 2025.
UF URS 2025	Arushi Pant, Jessica Jiang, Deepak Chhangani, **Shivam Kaushik**, Swapnil Pandey, and Diego E. Rincon-Limas, *Drosophila models of Amyotrophic lateral sclerosis (ALS)*, Undergraduate Research Symposium organized by University of Florida, Gainesville, Florida, USA. October 16, 2025.
IRCND 2025	**Shivam Kaushik**, Mayanglambam Dhruba Singh, Jean R. Geste, Vanlalrinchhani Varte, Swapnil Pandey, Deepak Chhangani, and Diego E. Rincon-Limas, *A Drosophila-based genetic screen reveals new modifiers of Abeta and Tau co-pathology*, 5th International Research Conference on Neurodegenerative Diseases (IRCND 2025), Gainesville, Florida, USA. July 9–12, 2025.
IRCND 2025	Swapnil Pandey, **Shivam Kaushik**, Deepak Chhangani, Yang S., Johnson A.N., and Diego E. Rincon-Limas, *Can SARS-Co-V2 Proteins Catalyze Abeta42-Induced Toxicity In Vivo?*, 5th International Research Conference on Neurodegenerative Diseases (IRCND 2025), Gainesville, Florida, USA. July 9–12, 2025.
AD/PD 2025	Swapnil Pandey, **Shivam Kaushik**, Deepak Chhangani, Yang S., Johnson A.N., and Diego E. Rincon-Limas, *Can SARS-COV-2 proteins accelerate Abeta-42-induced neurodegeneration IN VIVO?*, 19th International Conference on Alzheimer's & Parkinson's Diseases Online, Vienna, Austria. April 1–5, 2025.
Neurology Day 2025	**Shivam Kaushik**, Mayanglambam Dhruba Singh, Jean R. Geste, Vanlalrinchhani Varte, Swapnil Pandey, Deepak Chhangani, and Diego E. Rincon-Limas, *Discovering Novel Targets and Mechanisms Underlying Alzheimer Disease Pathogenesis in Drosophila*, Neurology Research Day held in Gainesville, Florida by Neurology Department, University of Florida. January 24, 2025.
Neurology Day 2025	Swapnil Pandey, Deepak Chhangani, **Shivam Kaushik**, and Diego E. Rincon-Limas, *The microRNA miR-33 regulates TDP-43-induced neurotoxicity in a fly model of FTLD/ALS*, Neurology Research Day held in Gainesville, Florida by Neurology Department, University of Florida. January 24, 2025.
ANA 2024	Swapnil Pandey, Deepak Chhangani, **Shivam Kaushik**, and Diego E. Rincon-Limas, *The microRNA miR-33 regulates TDP-43-induced neurotoxicity in a fly model of FTLD/ALS*, 149th American Neurological Association conference held in Orlando, Florida, USA. September 14–17, 2024.
TAGC 2024	Swapnil Pandey, Deepak Chhangani, **Shivam Kaushik**, and Diego E. Rincon-Limas, *A microRNA screening reveals miR-33 as a modifier of TDP-43 toxicity in flies*, TAGC-2024, held in Washington DC. March 6–10, 2024.
NCF SfN 2024	**Shivam Kaushik**, Mayanglambam Dhruba Singh, Jean R. Geste, Vanlalrinchhani Varte, Swapnil Pandey, Deepak Chhangani, and Diego E. Rincon-Limas, *Discovering Novel Targets and Mechanisms Underlying Alzheimer's Disease Pathogenesis in Drosophila*, 14th Annual NCF SfN Regional Conference, University of Florida, FL, USA. February 25–26, 2024.
TAGC 2023	Swapnil Pandey, Deepak Chhangani, **Shivam Kaushik**, and Diego E. Rincon-Limas, *The microRNA miR-33 regulates mutant TDP-43 toxicity in transgenic flies*, TAGC-2024, Chicago. March 1–5, 2023.

ICPMH 2021	**Shivam Kaushik**, Diego E. Rincon-Limas, and Pinky Kain, *Starvation and activity dependent modulation of salt taste behaviour in Drosophila*, International Conference on Psychology and Mental Health, Dubai, UAE. November 2021.
AAIC 2021	**Shivam Kaushik**, Diego E. Rincon-Limas, and Pinky Kain, *Starvation and activity dependent modulation of salt taste behaviour in Drosophila*, Alzheimer's Association International Conference Neuroscience Virtual. 2021.
IBRO-APRC 2021	**Shivam Kaushik** and Pinky Kain, *Understanding the role of taste circuits and its modulation using Drosophila melanogaster model*, Virtual IBRO-APRC Associate School of Neuroscience 2021 (**Best Oral Presentation Award**). 2021.
APDR5 2020	**Shivam Kaushik**, Rahul Kumar, Srishti Sanghi, Sachin Kumar, and Pinky Kain, *Starvation and neuronal activity in sensitized pharyngeal neurons induce modulation of salt taste behaviour in Drosophila melanogaster*, APDR5 Conference (IISER-PUNE). 2020.
SNCI 2019	**Shivam Kaushik**, Rahul Kumar, Srishti Sanghi, Sachin Kumar, and Pinky Kain, *Starvation and neuronal activity in sensitized pharyngeal neurons induce modulation of salt taste behaviour in Drosophila melanogaster*, SNCI 2019 Conference (Jamia Hamdard, New Delhi). 2019.
PAC 2019	**Shivam Kaushik**, Rahul Kumar, Srishti Sanghi, Sachin Kumar, and Pinky Kain, *Starvation and neuronal activity in sensitized pharyngeal neurons induce modulation of salt taste behaviour in Drosophila melanogaster*, PAC meeting of the Regional Centre for Biotechnology. 2019.

SKILLS

Drosophila Genetics

- **Genetic Workflows:** Extensive experience with Drosophila melanogaster as a model organism, including complete handling of genetic workflows and behavioral/physiological assays. Skilled in designing and executing complex genetic crosses, maintaining mutant and transgenic lines, and utilizing advanced genetic tools such as GAL4-UAS, RNAi knockdown, LexA/LexAop, and optogenetic systems.
- **Microdissections:** Proficient in larval and adult dissections (brain, heart, gut, labellum), sample preparation, phenotypic characterisation, and high-resolution imaging of tissues.
- **Feeding & Sensory Assays:** Experience performing Two-choice feeding assays, PER (Proboscis Extension Response), CAFE assay, and spectrophotometric quantification for metabolic/feeding-related studies.
- **Behavioral Paradigms:** Demonstrated expertise in behavioral assays, including locomotor performance, learning and memory paradigms (Fly-on-Ball memory assay, Zantiks climbing & memory assays), and sensory response profiling.
- **Tissue Analysis:** Adept in calcium imaging, Two Photon imaging, immunohistochemistry, and quantitative analysis of neural and peripheral tissues (brain, gut, VNC, heart, wings, legs, labellum).

Microscopy & Imaging

- **Advanced Microscopy:** Proficient in Two-photon calcium imaging for neuronal activity monitoring, confocal and super-resolution confocal microscopy for detailed subcellular visualization, and SEM for ultrastructural characterization.
- **Phenotypic Screening:** Competent in fluorescence and stereo microscopy for tissue-level imaging, developmental staging, and phenotypic screening.
- **Acquisition:** Strong understanding of image acquisition parameters, optimization, and analytical workflows.

Molecular Biology Techniques

- **Pathway Analysis:** PCR, DNA isolation, gel electrophoresis, experienced in Western blotting, immunoblotting, and protein–protein interaction analysis for evaluating molecular pathways.
- **Tissue Mapping:** Skilled in immunohistochemistry of complex Drosophila tissues (brain, gut, heart, VNC) with precise antibody-based localization and imaging.
- **Histology:** Familiar with preparing tissues for molecular and histological studies, including fixation, dissection, sectioning, and staining for high-resolution imaging.

Genetics (Classical & Molecular)

- **Core Genetics:** Genotype–phenotype correlation analysis, Maintaining and tracking multiple genetic lines, Applying molecular genetics approaches for functional gene analysis, Understanding inheritance patterns in Drosophila.

Bioinformatics Biology

- **Computational Analysis:** Proficient in computational tools for sequence and structure analysis. Experience with sequence alignment (BLAST, Clustal Omega, MUSCLE), protein–protein interaction prediction, structural modelling using AlphaFold2, and molecular visualization tools for structural interpretation.
- **Data Integration:** Familiar with analyzing biological databases (NCBI, UniProt, FlyBase, PDB) and integrating computational insights with wet lab findings to strengthen experimental conclusions.

Software & Data Analysis

- **Statistics & Languages:** MATLAB, GraphPad Prism, Origin, C++ (basic understanding), Biopython.
- **Scientific Illustration:** Adobe Illustrator, Photoshop, Premiere, ChemDraw, Microsoft Office Suite,

AWARDS & FELLOWSHIPS

- **Graduate Research Assistant**, *University of North Carolina, Chapel Hill, NC, USA*, 2022.
- **Presentation Award**, *Virtual IBRO-APRC Associate School of Neuroscience: Gut Microbiota and the Gut-Brain Connection*, 2021.

VOLUNTEER ACTIVITIES

Science Outreach & Leadership

- **Curriculum Development:** Designed and led the "We Are What We Eat" outreach program, focusing on nutrition and biological sciences for over 200 students. (Nov 2018)
- **Program Organizer:** Science outreach initiative focused on STEM education for secondary students at Government Sr. Sec. School, New Delhi. (July 2017)

Community Engagement

- **Outreach Member:** Organized and executed multi-site educational programs at Government Girls Inter College, Tehri, and Government Sr. Sec. School, Tehri, Uttarakhand, India. (Nov 22–23, 2018)
- **Educational Advocacy:** Promoted scientific literacy and health awareness in under-resourced regional schools through interactive workshops and seminars.